MINKUN KIM

Mobile: +49 152 06345028 **/ Email:** kimmk0810@gmail.com **/ Citizenship:** South Korean, permanent residence in Germany

PERSONAL STATEMENT

Proficient in Bayesian modeling and a wide range of statistical methods for prediction, uncertainty quantification, and more. Proficient in ETL and large scale data management. Possess a deep understanding of the underlying mathematical theories and derivations, and enjoying applying and customizing these frameworks to address complex quantitative challenges in energy finance and risk management. Additionally, interested in integrating ESG-related topic into financial risk management process and value generation.

Website: https://mainkoon81.github.io/minkun-website/

Main Tools: R, C++, Python
Financial Mathematics Skills: Stochastic Calculus, Matrix Calculus, Differential Equation, Time Series, Simulation
Data Engineering Skills: SQL, Databricks, GCP, **Apache**(Spark, Airflow, Kafka), Docker, Kubernetes, DBT, Tableau
Quantitative Financial Research Skills: Risk + Valuation modelling and Scientific paper writing
Sustainability Reporting: Taxonomy analysis, DMA, ESRS-E1/S1/G1 reporting
Language Skills: English (full professional proficiency), Korean (native), German (A1)

EDUCATION AND TRAINING

M.Sc Mathematical Finance + Sustainable Energy System Management — 2025-2028

School of Mathematics, University of York, UK

School of Economics, Ruhr University Bochum, Germany

Key Modules: Mathematical Methods in Finance, Discrete Time Modelling and Derivative Securities, Portfolio Theory and Risk Management, Stochastic Calculus and Black-Scholes Theory, Computational Finance, Modelling of Bonds+Terms Structure+Interest Rate Derivatives, Credit Risk Management, Energy Systems, Renewable Energy Technologies, Microeconomics, Climate Economics, Circular Economics, Energy Investments and Finance, Energy Regulation and Law, Business Models for the Energy Transition, Modern Project Management, Innovation and Technology Management, Implementing Renewable Energy Project, Application of Smart Energy, Accounting for Sustainability, etc.

[Thesis]: T.B.D

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Related Publications

1) **T.B.D**
   Journal
2) **T.B.D**
   Journal

Ph.D Computing with research focus on Bayesian Finance — 2020-2025

ADAPT Centre, Dublin City University, Ireland

Research Topic: Bayesian Parametric/Nonparametric Risk Modeling with Incomplete Data in Actuarial Practice

[Thesis]: Insurance Risk Premium Development with Model Risk, 2025; Available online at: (https://doras.dcu.ie/30936/)

Related Publications

1) **Dirichlet Process Log Skewnormal Mixture with Missing at Random Covariate in Insurance Practice**
   Journal Econometrics, 2023, 11(4), 24; Available online at: (https://www.mdpi.com/2225-1146/11/4/24)
2) **Bayesian Hierarchical Risk Premium Modeling with Model Risk: Addressing Non-Differential Berkson Error**
   Journal Applied Sciences, 2024, 15(1), 210; Available online at: (https://doi.org/10.3390/app15010210)

Institute and Faculty of Actuaries (IFoA), UK, Actuarial Qualification — 2022-present

Completed Exams: CS1-Actuarial Statistics I, CM1-Actuarial Mathematics I, CB2-Business Economics

H.Dip Statistics + M.Sc Data Analytics — 2017-2019

School of Statistics and Mathematics, University College Dublin, Ireland

School of Computing, Dublin City University, Ireland

[Thesis]: Study of walkability on house price change in Dublin with WLS regression, Google Map API and Tableau (Commissioned by Dublin City Council). Available online at: (https://arxiv.org/abs/2310.07563)

B.Sc Urban Engineering + Architecture **2006-2015**

School of Engineering, Hongik University, Seoul, South Korea

School of Architecture, Rice University, Houston, USA

CAREER EXPERIENCE

Research Intern **2025-2026**

t.b.d, Germany

- **project:** t.b.d

Research Assistant **2019-2020**

ARC-SYM (Advanced Research Computing for Complex Systems Modelling Centre), Ireland

- **Bioinformatics project:** Helped the physicians from St.Vincent University Hospital in Dublin, developing research w.r.t Gene Regulatory Network: (1) Dimensionality reduction, (2) Feature Selection: Modeling with Diverse Risk Factors/Determinants (genetic/clinical), (3) Refining GRS Risk Factors in the model; Integration between Traditional Risk Factors, Genetic Variants (SNP, GRS, etc.)

Intern **2014-2015**

Seoul Housing and Communities Corporation (SH Corporation), South Korea

- **Urban re-generation project:** Helped urban developers with site selection, acquisition, planning, space branding, and content producing, etc. I leveraged analytical skills (using ArcGIS) to pinpoint high-potential locations ripe for revitalization. This involved conducting thorough mathematical analyses to advocate for the inclusion of these areas in redevelopment plans.

Soldier **2008-2010**

Military Supply Specialist & Anti-Aircraft Gun (AAG) Operator, Air Force, Republic of Korea

OTHER INFORMATION

Extra-Curricular Activities and Interests

- **Freelance work 2019**– Helped small online shopping business examine Click Through Rate and Conversion Rate, trying 3 different A/B testing approaches such as (1) CI & Bootstrapping, (2) Simulating directly from the Null Hypothesis, (3) Regression: OLS, Logistic, KNN, SVM.
- **International MediaEval Competition 2018** – Achieved 45% accuracy from my MLP Structure that I designed to predict memorability scores for videos, which reflect the probability of a video being remembered by majority of people. Using an Autoencoder for Greedy Layer-Wise Pre-Training and Learning Rate Scheduling for SGD. Available online at (https://www.github.com/mainkoon81).
- **Twitter better Rating Algorithm with Computer Vision 2017** – Developed tweeting recognition algorithm as part of a web application for dog lovers. Web scrapping and investigating Twitter users' retweeting behavior, I first brought up an insight on how to get more retweets and favorites. This is followed by my tweeting recognition algorithm by taking any user-supplied dog image as input for classification then providing an estimate of the dog's breed to get more retweets and favorites.
- **Recipient of a prize at the Korea Architecture Competition 2010** – Applied Chora's *Liminal Body* concept to the redevelopment of a major commercial street in Jung-gu, Seoul, exploring spatial transitions and urban connectivity and integrating dynamic public spaces with commercial infrastructure.

Hobbies: Tutoring mathematics, and exploring spatial design driven by economic forces and risk dynamics.